

HW #05

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F

(Q:01)

$$B = \{e^x, xe^x, x^2e^x\}$$

$$D(e^x) = e^x$$

$$D(xe^x) = xe^x + e^x$$

$$D(x^2e^x) = x^2e^x + 2xe^x$$

$$[D(e^x)]_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$[D(xe^x)]_B = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$[D(x^2e^x)]_B = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

$$[D]_B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

(b)

$$D \cdot D = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$D^2 \cdot D^2 = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 & 12 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix}$$

$$x^2 e^x + 2x e^x$$

$$\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} + 2 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}$$

fourth derivative :-

$$\begin{bmatrix} 1 & 4 & 12 \\ 0 & 1 & 8 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 20 \\ 10 \\ 1 \end{bmatrix}$$

$$20x^2 e^x + 10x e^x + x^2 e^x$$

(c)

inverse :-

$$\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -1 & 2 \\ 0 & 1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 0 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 1 & 3 \end{array} \right] \begin{bmatrix} 0 \\ -1 \\ 3 \end{bmatrix} = \begin{bmatrix} 7 \\ -7 \\ 3 \end{bmatrix}$$

(Q: 2)

$$D(\sin x) = \cos x$$

$$D(\cos x) = -\sin x$$

$$\begin{bmatrix} \cos x & 0 \\ 0 & -\sin x \end{bmatrix} [D]_B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = 0$$

$$\lambda^2 + 1 = 0$$

$$\lambda = \pm i$$

$$\lambda = i$$

$$x = \begin{bmatrix} i \\ i \\ 1 \end{bmatrix}$$

$$\lambda = -i$$

$$x = \begin{bmatrix} -i \\ -i \\ 1 \end{bmatrix}$$

$$A^{1/2} = P D^{1/2} P^{-1}$$

$$A^{1/2} = \begin{bmatrix} i & -i \\ 1 & 1 \end{bmatrix} \begin{bmatrix} i^{1/2} & 0 \\ 0 & -i^{1/2} \end{bmatrix} \begin{bmatrix} 1/2i & 1/2 \\ -1/2i & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} i^{3/2} & i^{3/2} \\ i^{1/2} & -i^{1/2} \end{bmatrix} \begin{bmatrix} 1/2i & 1/2 \\ -1/2i & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & -1 \\ +i & -i \end{bmatrix} \begin{bmatrix} 1/2i & 1/2 \\ -1/2i & 1/2 \end{bmatrix}$$

$$= \begin{bmatrix} -1/2i + 1/2i & -1/2 - 1/2 \\ +1/2i + 1/2i & 0 \end{bmatrix}$$

$$A^{1/2} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$